

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-11 19:30 UTC

SEC P vs NP — notebook summary

Generated 2026-05-11 19:30 UTC

- Cycles recorded: **619**
- Time span: 429.8h (~1.44 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	582
BARRIER_HIT	18
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	34
Schur-Weyl Duality	19
Representation Theory of Symmetric Groups	17
Free Probability	14
Matroid Theory	12
Additive Combinatorics	10
Noncommutative L^p Geometry	8
Polymatroid Theory	8
Algebraic Geometry	7
Random Matrix Theory	7
Finite Geometry	6
Spectral Graph Theory	6
Plethysm Theory	6
Free Probability Theory	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4

Field	Cycles
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3
Diophantine Approximation	3
Persistent Homology	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-11 13:48 UTC	INCONCLUSIVE	Secant Rank of Disjointness Communication Matrix Bounds Randomize
2026-05-11 14:19 UTC	INCONCLUSIVE	Kronecker Coefficient Gap in Symmetric Powers of Permanent vs Det
2026-05-11 14:27 UTC	INCONCLUSIVE	Non-Commutative Rank Gap in Read-Twice BPs for IP_2
2026-05-11 14:33 UTC	INCONCLUSIVE	Standard Young Tableaux Count vs Monotone Circuit Size
2026-05-11 14:42 UTC	INCONCLUSIVE	Real Rank of Moment Matrix Bounded by SOS Refutation Degree for C
2026-05-11 14:59 UTC	INCONCLUSIVE	Schatten p-Norm Inverse Proportional to Disjointness Communicatio
2026-05-11 15:18 UTC	INCONCLUSIVE	Tree-Depth Exponentiation Bounds Tseitin Resolution Length
2026-05-11 15:41 UTC	INCONCLUSIVE	Resultant Degree Bounds Frege Proof Size for Tautologies
2026-05-11 15:49 UTC	INCONCLUSIVE	Plethysm Coefficient Gap in Monotone Circuit Complexity for Perma
2026-05-11 16:11 UTC	INCONCLUSIVE	Real Rank of Depth-3 AC ⁰ Circuits for PARITY is $\Omega(\log n)$
2026-05-11 17:00 UTC	INCONCLUSIVE	Quadratic Form Solution Count Exponential Gap in ACC ⁰ Sipser Circ
2026-05-11 17:52 UTC	INCONCLUSIVE	Additive Energy Inverse Proportional to Discrepancy of Sumset Com

Time	Verdict	Title
2026-05-11 18:06 UTC	INCONCLUSIVE	Symmetric Square Multiplicity Gap in Permanent vs Determinant Dec
2026-05-11 18:31 UTC	INCONCLUSIVE	Free Cumulant Sum Gap in Read-Twice BP Transition Matrices
2026-05-11 19:29 UTC	INCONCLUSIVE	Colin de Verdière Invariant Lower Bounds Resolution Length for Ts

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (sudo apt install pandoc texlive-xetex), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```